

BODAYBO, Russian FederationWMO: **302530**Lat: **57.865N**Lon: **114.239E**Elev: **920**StdP: **14.21**Time Zone: **8.00 (E08)**Period: **94-05**WBAN: **99999****Annual Heating, Humidification, and Ventilation Design Conditions**

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-47.3	-44.0	-50.8	0.3	-47.1	-47.8	0.3	-43.9	12.5	2.3	9.9	3.3	0.1	200	0.657

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	22.7	86.6	65.8	82.8	64.9	79.1	63.4	68.7	80.0	66.9	78.1	65.3	75.8	4.8	50

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
65.5	97.3	71.7	63.2	89.6	70.8	61.1	83.2	70.1	33.5	81.5	32.1	77.6	30.7	76.5	78.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 13.4	11.2	9.3	DB	-49.3	92.9	2.7	1.0	-51.2	93.6	-52.8	94.2	-54.3	94.8	-56.2	95.5	(4)
(5)			WB	-49.6	71.6	2.7	3.5	-51.5	74.1	-53.1	76.2	-54.6	78.1	-56.6	80.6	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	24.7	-20.6	-8.4	8.4	29.5	44.4	59.4	67.4	61.1	44.6	26.5	0.9	-18.8		
	DBStd	32.38	14.46	15.29	13.40	7.48	7.67	7.50	6.00	5.98	7.05	9.77	16.60	15.59		
	HDD50	10472	2188	1634	1291	615	209	13	1	2	188	727	1472	2132		
	HDD65	14878	2653	2054	1756	1065	638	195	45	149	612	1192	1922	2597		
	CDD50	1244	0	0	0	0	36	296	541	345	26	0	0	0		
	CDD65	175	0	0	0	0	0	28	120	27	0	0	0	0		
	CDH74	2079	0	0	0	0	31	490	1268	288	2	0	0	0		
	CDH80	625	0	0	0	0	4	137	421	63	0	0	0	0		
(14) Wind	WSAvg	3.2	1.5	2.4	3.5	4.6	4.5	3.9	3.3	3.0	3.3	3.5	2.6	1.6		
(15) Precipitation	PrecAvg	17.6	0.9	0.5	0.4	0.6	1.5	2.6	3.0	3.0	2.3	0.9	1.1	1.0		
	PrecMax	23.5	1.7	1.1	1.0	1.5	3.6	5.1	7.2	6.1	5.1	2.0	1.9	1.8		
	PrecMin	13.1	0.2	0.2	0.1	0.1	0.3	0.1	0.7	0.8	1.0	0.2	0.4	0.4		
	PrecStd	3.0	0.3	0.2	0.2	0.3	0.7	1.0	1.6	1.5	0.9	0.4	0.5	0.3		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	20.0	30.0	38.6	57.5	78.4	89.0	91.3	86.6	70.1	47.5	38.2	22.5		
		MCWB	18.1	24.8	31.5	43.6	57.4	64.1	67.6	66.0	57.7	39.1	34.7	19.9		
	2%	DB	10.4	23.7	34.3	49.5	71.6	84.2	88.2	80.7	64.0	43.6	33.9	17.3		
		MCWB	9.1	21.2	28.8	39.7	53.2	62.8	66.8	63.4	53.2	36.7	31.7	15.8		
	5%	DB	4.3	18.9	31.2	44.9	66.5	80.1	84.8	77.2	59.1	40.5	30.2	10.0		
		MCWB	3.5	17.1	26.7	36.2	50.6	61.6	65.8	63.8	51.3	35.9	28.3	8.9		
	10%	DB	-0.1	13.5	27.8	41.0	60.3	75.6	81.4	73.3	55.1	37.9	23.2	4.4		
		MCWB	-0.8	12.0	23.8	33.8	47.9	59.9	65.2	62.0	49.7	34.4	21.4	3.6		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	18.5	25.1	32.7	43.9	59.3	67.3	71.3	70.2	60.2	40.9	35.2	20.5		
		MCDB	20.5	29.2	36.5	54.0	77.8	80.9	84.6	77.3	67.7	43.8	37.2	22.0		
	2%	WB	9.0	21.6	29.4	40.1	54.7	65.2	69.3	67.2	55.0	38.4	32.1	15.4		
		MCDB	10.4	23.8	34.6	48.8	69.0	78.5	81.6	75.0	60.9	41.5	33.7	17.5		
	5%	WB	3.6	16.9	26.5	37.0	51.7	63.6	68.1	65.2	52.5	36.5	28.1	9.0		
		MCDB	4.4	18.8	31.1	43.3	65.5	76.0	80.1	74.0	57.5	39.5	30.3	10.2		
	10%	WB	-0.7	12.2	23.8	34.5	48.4	61.6	66.7	63.2	50.1	34.8	21.4	3.7		
		MCDB	0.1	13.7	27.4	40.2	58.8	73.0	78.2	71.6	54.6	37.4	23.1	4.5		
(35) Mean Daily Temperature Range	MDBR		12.0	16.8	22.5	18.8	21.5	23.4	22.7	20.2	15.0	11.7	12.5	11.7		
	5% DB	MCDBR	15.9	17.7	19.7	24.0	34.2	32.7	31.8	28.9	23.8	15.1	11.9	15.5		
		MCWBR	15.2	15.1	15.1	15.3	17.8	14.0	12.5	13.0	13.7	10.2	9.6	14.3		
	5% WB	MCDBR	15.8	17.2	18.9	22.4	32.0	26.9	25.5	24.8	21.1	13.4	11.6	15.3		
		MCWBR	15.1	14.7	14.6	14.9	18.0	13.2	11.2	12.0	13.1	10.0	9.9	14.2		
(40) Clear-Sky Solar Irradiance	taub		0.256	0.292	0.315	0.357	0.403	0.436	0.447	0.403	0.342	0.300	0.263	0.240		
	taud		2.005	2.013	2.042	2.052	2.152	2.217	2.219	2.326	2.436	2.330	2.047	1.964		
	Ebn at Noon		179	227	259	267	263	255	248	252	254	231	173	144		
	Edh at Noon		18	29	38	44	43	41	40	34	26	21	17	14		
(44) All-Sky Solar Radiation	RadAvg		215	517	1049	1602	1729	1703	1584	1276	777	426	248	141		
	RadStd		14	30	52	76	134	146	171	111	82	51	30	11		

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD50	HDD65	CDD50	CDD65
(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)			
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (2 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page